

Matplotlib



What is Matplotlib?

Matplotlib is a Python library used for:

- Data visualization
- Creating charts and graphs
- Plotting 2D data

It works very well with **NumPy** and **Pandas**.

The most commonly used module is:

```
import matplotlib.pyplot as plt
```



Installation

```
pip install matplotlib
```



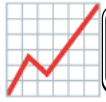
Basic Plot Structure

```
import matplotlib.pyplot as plt
```

```
x = [1,2,3,4]  
y = [10,20,25,30]
```

```
plt.plot(x, y)  
plt.show()
```

`plt.show()` → displays the graph.



1 Line Plot

```
plt.plot(x, y)
plt.title("Line Chart")
plt.xlabel("X Axis")
plt.ylabel("Y Axis")
plt.show()
```

Important `plot()` parameters

- `color`
- `linestyle`
- `linewidth`
- `marker`
- `label`

Example:

```
plt.plot(x, y, color='red', linestyle='--', marker='o', label='Data')
plt.legend()
```



2 Bar Chart

```
plt.bar(x, y)
plt.show()
```

Horizontal bar:

```
plt.barh(x, y)
```



3 Pie Chart

```
labels = ['A','B','C']
sizes = [40, 35, 25]
```

```
plt.pie(sizes, labels=labels, autopct='%1.1f%%')  
plt.show()
```



4 Histogram

```
data = [1,2,2,3,3,3,4,5]
```

```
plt.hist(data, bins=5)  
plt.show()
```

Used for frequency distribution.



5 Scatter Plot

```
plt.scatter(x, y)  
plt.show()
```



6 Subplots (Multiple Graphs)

```
plt.subplot(1,2,1)  
plt.plot(x,y)
```

```
plt.subplot(1,2,2)  
plt.bar(x,y)
```

```
plt.show()
```

Better way:

```
fig, ax = plt.subplots(2,2)
```



Customization Functions

Function	Purpose
<code>plt.title()</code>	Add title
<code>plt.xlabel()</code>	X-axis label
<code>plt.ylabel()</code>	Y-axis label
<code>plt.legend()</code>	Show legend
<code>plt.grid()</code>	Add grid
<code>plt.xlim()</code>	X-axis limit
<code>plt.ylim()</code>	Y-axis limit
<code>plt.xticks()</code>	Set x ticks
<code>plt.yticks()</code>	Set y ticks
<code>plt.style.use()</code>	Change style

Example:

```
plt.grid(True)
plt.style.use('ggplot')
```



Saving Figures

```
plt.savefig("chart.png")
```



Figure Size

```
plt.figure(figsize=(8,5))
```

Object-Oriented Method (Recommended)

```
fig, ax = plt.subplots()
```

```
ax.plot(x,y)  
ax.set_title("Line Plot")  
ax.set_xlabel("X")  
ax.set_ylabel("Y")
```

```
plt.show()
```

Complete List of Common Matplotlib Functions

◆ Plotting Functions

- `plt.plot()`
- `plt.bar()`
- `plt.barh()`
- `plt.hist()`
- `plt.scatter()`
- `plt.pie()`
- `plt.boxplot()`
- `plt.violinplot()`
- `plt.stackplot()`
- `plt.step()`
- `plt.fill_between()`
- `plt.errorbar()`

◆ Figure & Axis Control

- `plt.figure()`
- `plt.subplots()`
- `plt.subplot()`
- `plt.gca()`
- `plt.gcf()`
- `plt.axes()`

◆ Labels & Text

- `plt.title()`
- `plt.xlabel()`
- `plt.ylabel()`
- `plt.legend()`
- `plt.text()`
- `plt.annotate()`

◆ Axis Control

- `plt.xlim()`
- `plt.ylim()`
- `plt.xticks()`
- `plt.yticks()`
- `plt.axis()`
- `plt.grid()`

◆ Styling

- `plt.style.use()`
- `plt.rcParams`

- `plt.setp()`
-

◆ Saving & Display

- `plt.show()`
 - `plt.savefig()`
 - `plt.close()`
-



Important Concepts

1 Figure

Whole chart area

2 Axes

Actual plotting area

3 Backend

Engine that displays plots (Jupyter, Tkinter, etc.)



Matplotlib with Pandas

```
import pandas as pd
```

```
df = pd.DataFrame({  
    "x": [1, 2, 3, 4],  
    "y": [10, 20, 15, 25]  
})
```

```
df.plot(x="x", y="y")  
plt.show()
```



Most Used Charts in Data Science

- Line plot → Trends
 - Bar chart → Comparison
 - Histogram → Distribution
 - Scatter plot → Relationship
 - Box plot → Outliers
-



Beginner Learning Order

1. Line plot
 2. Bar chart
 3. Histogram
 4. Scatter plot
 5. Subplots
 6. Customization
 7. Object-oriented approach
-

If you want next:

- ☒ Seaborn complete notes
- ☒ 50 matplotlib practice questions
- ☒ Real data visualization project
- ☒ Advanced matplotlib (3D plots, animations)
- ☒ Interview questions

Tell me what you want 